FSKU: Supplementary Material

Evaluating Financial Security Knowledge of LLMs on a Domain Expert-Validated Question-Answering Benchmark

S1. Short-form QA results

Domain / sub-domain	GPT-3.5	GPT-5	Qwen3: 14b	Qwen3: 32b	Deepseek -r1: 14b	Deepseek -r1: 32b	Claude-4	Gemma3: 27b	Gemini-Pr o	Qwen2.5	Llama3.1
GENERAL FINANCIAL SECURITY											
Financial business	55.56	77.78	55.56	66.67	55.56	55.56	77.78	66.67	77.78	55.56	44.44
Security governance	55.56	77.78	55.56	66.67	55.56	66.67	77.78	66.67	77.78	55.56	44.44
Electronic security	55.56	77.78	66.67	66.67	55.56	66.67	77.78	66.67	77.78	55.56	44.44
Privacy	55.56	66.67	55.56	55.56	44.44	55.56	77.78	66.67	66.67	44.44	33.33
FINANCE IT SECURITY											
General IT security	55.56	77.78	66.67	66.67	55.56	66.67	77.78	66.67	66.67	55.56	33.33
Advanced IT security	66.67	77.78	66.67	77.78	66.67	66.67	88.89	77.78	77.78	66.67	44.44
FINANCE LEGAL & REGULATION											
Electronic Finance	44.44	55.56	44.44	44.44	33.33	44.44	66.67	44.44	55.56	33.33	22.22
Telecommunications	44.44	66.67	44.44	55.56	33.33	44.44	66.67	55.56	66.67	33.33	22.22
Privacy Law	33.33	55.56	44.44	44.44	33.33	44.44	66.67	44.44	55.56	33.33	22.22
Digital Signature	33.33	66.67	44.44	55.56	44.44	55.56	77.78	55.56	55.56	33.33	33.33
Credit Information	44.44	66.67	44.44	55.56	44.44	55.56	66.67	55.56	55.56	33.33	22.22
EMERGING TRENDS											
Financial Security Trends	55.56	66.67	66.67	66.67	55.56	55.56	77.78	66.67	66.67	44.44	44.44
Average scores	50.00	69.44	54.63	60.19	48.15	56.48	75.00	61.11	66.67	45.37	34.26

Table S1. Overall performance (%) across domains and sub-domains on FSKU short-form QA. Original rank highlighting is retained: red = 1st, orange = 2nd, yellow = 3rd.

S2. Dataset sources and benchmark construction

Reference sources by domain and sub-domain

Table S2 summarizes the reference materials used to construct FSKU across the four financial security domains and twelve sub-domains. The percentage shown for each sub-domain indicates the proportion of QA items allocated in the final benchmark. The distribution was determined to provide balanced coverage while reflecting the practical importance of each topic in financial security. Official certification textbooks were mainly used for general financial security and advanced IT security topics, whereas laws, supervisory regulations, and security guidelines were adopted for finance legal & regulation and privacy-related sub-domains.

Domain	Sub-domain / ratio	Reference material
General Financial Security	Financial business (5.0%)	FSM textbook Vol.1 (Finance Business)
	Security governance (5.0%)	FSM textbook Vol.2 (Security Management System)
	Electronic security (5.0%)	FSM textbook Vol.4 (E-financial security)
	Privacy (5.0%)	ISMS-P certification guidelines
Finance IT security	General IT security (12.0%)	Information security engineer exam guidelines
	Advanced IT security (8.0%)	FSM textbook Vol.3 (Advanced IT Security)
Finance legal & regulation	Electronic Finance (12.0%)	Electronic Financial Transactions Act
	Telecommunications (12.0%)	Information and Communications Network Act & Enforcement Decree
	Privacy Law (12.0%)	Personal Information Protection Act & Enforcement Decree
	Digital Signature (7.0%)	Digital Signature Act & Enforcement Decree
	Credit Information (12.0%)	Credit Information Use and Protection Act & Supervision Rules
Emerging Trends	Financial Security Trends (5.0%)	2024 Digital Finance and Cybersecurity Issues Outlook

Table S2. FSKU reference sources by domain and sub-domain. Ratios reproduce the allocation percentages reported in the source material.

Benchmark construction process

Figure S1 illustrates the overall pipeline for constructing FSKU. Starting from authoritative source materials, three certified financial security experts selected globally applicable documents covering the target knowledge areas. The selected documents were converted into a retrievable knowledge base using embedding models and a vector database. GPT-4o then generated candidate QA items, guided by predefined generation rules and prompt instructions. Finally, all generated questions were reviewed and refined by financial security experts according to the eight quality criteria described in Section 4 of the main paper, followed by bilingual translation and manual verification. This human-in-the-loop pipeline ensures that every QA item is grounded in authoritative financial security knowledge while maintaining consistency and quality.

S2. Source-material illustrations

Construction workflow

Figure S2 presents representative examples of the authoritative materials used during benchmark construction. Figure S2(a) shows the official Financial Security Manager (FSM) qualification operated by the Financial Security Institute (FSI), which provides the curriculum and examination scope for financial security professionals. Figure S2(b) presents the examination criteria of the national Information Security Engineer qualification, which was used as a reference for IT security knowledge. Figure S2(c) shows the original legal text of the Credit Information Use and Protection Act, one of the primary references for constructing finance legal & regulation questions. These materials served as the factual foundation for generating domain-grounded QA items.

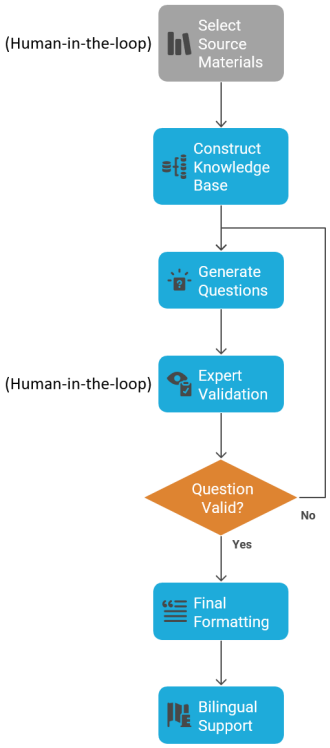
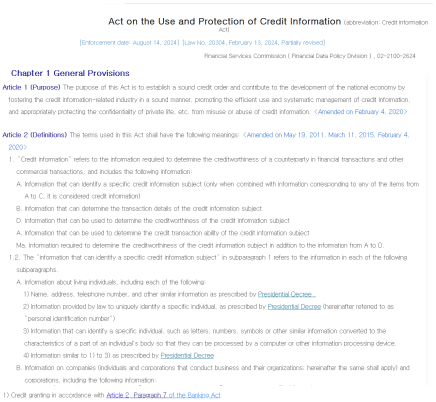


Figure S1. FSKU construction process.

(a) Financial Security Manager qualification


Act on the Use and Protection of Credit Information (abbreviation: Credit Information Act)

[Enforcement date: August 14, 2024] [Law No. 20304, February 13, 2024, Partially revised]

Financial Services Commission (Financial Data Policy Division) , 02-2100-2624

Chapter 1 General Provisions

☐ Article 1 (Purpose) The purpose of this Act is to establish a sound credit order and contribute to the development of the national economy by fostering the credit information-related industry in a sound manner, promoting the efficient use and systematic management of credit information, and appropriately protecting the confidentiality of private life, etc. from misuse or abuse of credit information. <Amended on February 4, 2020>

☐ Article 2 (Definitions) The terms used in this Act shall have the following meanings: <Amended on May 19, 2011, March 11, 2015, February 4, 2020>

1. "Credit information" refers to the information required to determine the creditworthiness of a counterparty in financial transactions and other commercial transactions, and includes the following information:

A. Information that can identify a specific credit information subject (only when combined with information corresponding to any of the items from A to C, it is considered credit information)

B. Information that can determine the transaction details of the credit information subject

C. Information that can be used to determine the creditworthiness of the credit information subject

A. Information that can be used to determine the credit transaction ability of the credit information subject

Ma. Information required to determine the creditworthiness of the credit information subject in addition to the information from A to D.

1.2. The "information that can identify a specific credit information subject" in subparagraph 1 refers to the information in each of the following subparagraphs.

A. Information about living individuals, including each of the following:

1) Name, address, telephone number, and other similar information as prescribed by Presidential Decree ;

2) Information provided by law to uniquely identify a specific individual, as prescribed by Presidential Decree (hereinafter referred to as "personal identification number");

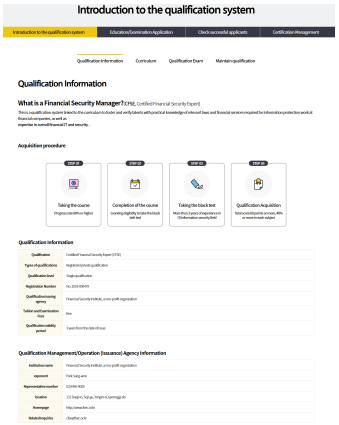
3) Information that can identify a specific individual, such as letters, numbers, symbols or other similar information converted to the characteristics of a part of an individual's body so that they can be processed by a computer or other information processing device;

4) Information similar to 1) to 3) as prescribed by Presidential Decree

B. Information on companies (individuals and corporations that conduct business and their organizations; hereinafter the same shall apply) and corporations, including the following information:

1) Credit granting in accordance with Article 2, Paragraph 7 of the Banking Act

(b) Information Security Engineer exam


Introduction to the qualification system

Introduction to the qualification system | Education/Examination Application | Check successful applicants | Certification Management

Qualification Information | Curriculum | Qualification Exam | Maintain qualification

Qualification Information

What is a Financial Security Manager? (CFSE, Certified Financial Security Expert)

This is a qualification system aimed to the curriculum to foster and verify talents with practical knowledge of relevant laws and financial services required for information protection work at financial companies, as well as expertise in overall financial IT and security.

Acquisition procedure

STEP 01 Taking the course — Progress rate 80% or higher

STEP 02 Completion of the course — Granting eligibility to take the Mock test/ test

STEP 03 Taking the Mock test — More than 2 years of experience in IT/Information security field

STEP 04 Qualification Acquisition — Status as a Reports at most, 40% or more mock subject

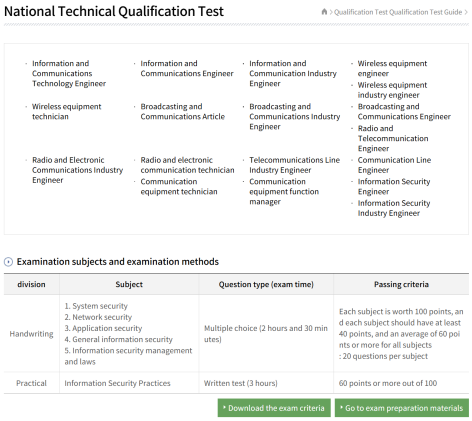
Qualification Information

Qualification	Certified Financial Security Expert (CFSE)
Types of qualifications	Registered private qualification
Qualification level	Single qualification
Registration Number	No. 2013-039415
Qualification issuing agency	Financial Security Institute, a non-profit organization
Tuition and Examination Fees	free
Qualification validity period	3 years from the date of issue

Qualification Management/Operation (Issuance) Agency Information

Institution name	Financial Security Institute, a non-profit organization
Location	(13) Yangjae-ro, Seocho-gu, Seoul
Homepage	http://www.fsec.or.kr
Website Inquiries	Cloud Plus 1:1

(c) Credit Information Use and Protection Act


National Technical Qualification Test

> Qualification Test Qualification Test Guide >

- Information and Communications Technology Engineer
- Information and Communications Engineer
- Information and Communication Industry Engineer
- Wireless equipment engineer
- Wireless equipment industry engineer
- Wireless equipment technician
- Broadcasting and Communications Article
- Broadcasting and Communications Industry Engineer
- Broadcasting and Communications Engineer
- Radio and Telecommunication Engineer
- Radio and Electronic Communications Industry Engineer
- Radio and electronic communication technician
- Communication equipment technician
- Telecommunications Line Industry Engineer
- Communication equipment function manager
- Communication Line Engineer
- Information Security Engineer
- Information Security Industry Engineer

Examination subjects and examination methods

division	Subject	Question type (exam time)	Passing criteria
Handwriting	1. System security 2. Network security 3. Application security 4. General information security 5. Information security management and laws	Multiple choice (2 hours and 30 minutes)	Each subject is worth 100 points, and each subject should have at least 40 points, and an average of 60 points or more for all subjects : 20 questions per subject
Practical	Information Security Practices	Written test (3 hours)	60 points or more out of 100

* Download the exam criteria | * Go to exam preparation materials

Figure S2. Examples of authoritative materials: (a) the FSI qualification and examination scope; (b) national Information Security Engineer examination criteria; (c) the original Credit Information Act text. Figures are reproduced from the accompanying manuscript.

S3. The 2025 Financial AI Challenge

As shown in Figure S3, the 2025 Financial AI Challenge is a national competition jointly organized by the Financial Security Institute (FSI), the Financial Services Commission, and five major financial institutions: Hana Bank, Shinhan Bank, KakaoBank, Mirae Asset Securities, and Shinhan Card. The competition aims to promote the adoption of trustworthy AI in the financial sector and to discover innovative financial service applications utilizing generative AI technologies. The challenge consists of two independent tracks:

Track I – Model Competition: 1,039 participants develop and evaluate generative AI models on the FSKU (Financial Security Knowledge Understanding) benchmark, a QA dataset designed by FSI to assess factual accuracy and domain knowledge relevance of AI responses in financial security contexts. As shown in Figure S4, Real-time ranking is provided via a leaderboard to enhance competitiveness and transparency (In this competition, participants were restricted to using only open-source models below 20B parameters.).

Track II – Service Ideation: Open to all participants, this track solicits creative ideas for AI-based financial customer support services tailored to user types (e.g., credit card users) and service channels (e.g., mobile apps).

The competition period spans from July 14 to August 29, 2025, with final awards presented during the FISCON 2025 conference in November 20. A total of KRW 33 million in prizes is awarded, including the Financial Services Commission Chairman's Award (KRW 15 million) for the grand prize winner. All winners receive preferential consideration for employment at FSI. Our experimental benchmark setup followed the same evaluation principles as the Challenge's Track I model competition, leveraging the FSKU dataset to measure finance security domain-specific accuracy and factual reliability.



Figure S3. Official competition poster.

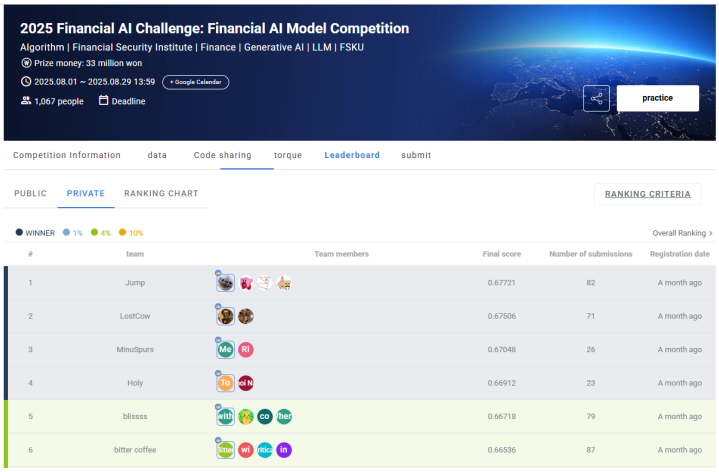


Figure S4. Leaderboard for Track I of the 2025 Financial AI Challenge.

S4. FSKU-Decision: difficulty and case study

The difficulty breakdown further illuminates the limits of current LLM capabilities in high-stakes governance settings. On the Hard cases (92 cases), top models maintain moderate proficiency (e.g., Claude-4 at 61.02 and GPT-5 at 59.72). However, performance sharply declines on the Expert cases (58 cases), where Claude-4 drops to 49.48 and GPT-5 to 47.25. The Expert subset requires identifying false premises, calibrating safe abstention, or performing multi-hop reasoning over conflicting logs. Most open-source LLMs score below 35.00 on Expert cases, frequently hallucinating mandatory regulatory filings even when internal logs explicitly refute an actual data breach (as illustrated in Table S3). These findings demonstrate that standard factual knowledge retention is insufficient for autonomous financial security decision-making, emphasizing the imperative for precise evidence grounding and calibrated uncertainty estimation.

Table S3 shows a false-premise case in which no incident is confirmed; here, the ground truth also specifies forbidden actions that an LLM must avoid recommending (e.g., escalating to authorities or notifying customers) despite surface-level cues suggestive of a breach, directly reflecting the required/acceptable/forbidden action taxonomy described in the main paper.

Field	Content
Domain	General financial security
Sub-domain	Electronic security
Task family	Incident triage (Expert)
Scenario	A suspected customer personal data leak was reported at a payment gateway. The reporter claims that, at 3 PM on the incident date, their payment details were exposed externally. Internal logs, external threat intelligence, and policy documents were collected.
Evidence	**E1:** Between 14:00 and 16:00 on the incident date, no abnormal access or external transmission records of customer payment data were found in internal systems. **E2:** No personal data breach alerts or malicious activity reports related to the gateway were found in threat intelligence feeds over the past month. **E3:** The relevant financial supervisory regulation mandates incident response activation upon breach signs, but prohibits definitive incident claims without clear evidence. **E4:** The reporter estimated payment details exposure but could not provide concrete evidence or logs. **E5:** An unrelated unauthorized login attempt from an external IP was detected and blocked; it was unconnected to customer payment data access.
Question	Can it be determined if this is an actual personal data breach incident? Provide the risk level, appropriate diagnosis codes, and response actions.
Answerability / Risk	False premise / Low
Diagnosis	NO_CONFIRMED_INCIDENT
Required actions	CONTINUE_MONITORING → DO_NOT_MAKE_DEFINITIVE_LEGAL_CLAIM → REQUEST_ADDITIONAL_INFORMATION
Forbidden actions	ACTIVATE_INCIDENT_RESPONSE CONTACT_AFFECTED_CUSTOMERS REPORT_TO_AUTHORITY
Required citations	E1, E2, E3, E4
Missing information	Concrete evidence of data exposure and verification of external transmission logs.
Reference rationale	Internal logs and external threat intelligence contradict the leak claim, and the reporter provides no concrete evidence. Because the regulation prohibits definitive incident claims without clear proof, escalating to incident response or notifying customers and authorities would be an unsupported and disruptive overreaction; the model should instead request additional information and continue monitoring.

Table S3. False-premise case with forbidden actions (FSKUD-0013; Expert). The reference labels, required actions, and forbidden actions reproduce the benchmark specification.

S5. Component-level evaluation and diagnostics

To provide a fine-grained diagnostic breakdown of model behavior on FSKU-Decision, Table S4 reports performance across the six evaluation components. In addition, we separately track JSON Validity and Safety Violation Rate to isolate formatting compliance and hazardous overreach from the aggregated functional scores. Figure S5 visualizes the results.

Model	Ans. Acc.	Risk Acc.	Diag. F1	Action quality	Evid. F1	Missing coverage	JSON validity	Safety viol. ↓
GPT-3.5	42.13	46.28	38.54	35.79	32.11	33.65	94.20	18.53
GPT-5	61.24	64.52	57.13	52.84	49.12	44.68	99.80	3.21
Qwen3:14b	39.47	43.82	35.18	33.12	30.57	31.22	96.80	16.42
Qwen3:32b	48.21	51.74	43.08	39.53	35.82	34.06	97.50	11.38
Deepseek-r1:14b	36.42	40.13	32.76	30.21	27.54	27.58	93.50	21.17
Deepseek-r1:32b	43.78	46.24	39.12	35.68	32.41	32.05	96.10	14.23
Claude-4	62.83	66.12	56.81	54.58	51.24	43.87	100.00	2.14
Gemma3:27b	47.12	50.38	42.03	38.61	34.89	34.88	98.20	12.76
Gemini-Pro	51.38	54.76	46.21	42.67	38.52	38.61	99.10	7.48
Qwen2.5	32.41	36.78	28.49	26.12	24.18	25.32	88.60	26.83
Llama3.1	18.23	21.48	15.12	12.38	9.82	8.47	82.40	34.79

Table S4. Component-wise scores (%) across 11 LLMs. Red, orange, and yellow indicate the three best distinct values per metric. Higher is better except for Safety Violation Rate (↓).

Reading the diagnostics

Ans. Acc. = answerability accuracy; Risk Acc. = risk-level accuracy; Diag. F1 = diagnosis F1; Action quality combines action-set F1 and order-aware nDCG; Evid. F1 = evidence-selection F1; Missing coverage = missing-information coverage. JSON validity and safety violations are reported separately from the six functional components.

As shown in Table S4, LLMs demonstrate significant structural disparities across different decision fields:

Categorical vs. Ranked Capabilities: Top LLMs such as Claude-4 and GPT-5 perform relatively well on discrete classification fields such as answerability (62.83% and 61.24%) and risk_level (66.12% and 64.52%). However, scores drop significantly on sequential and citation tasks, with Action Quality peaking at 54.58% (Claude-4) and Evidence Citation F1 at 51.24% (Claude-4). This gap indicates that while LLMs can diagnose an incident type, ordering containment actions and anchoring them to local evidence IDs remains considerably harder.

Identification of Missing Knowledge: Missing-Information Coverage scores are universally low across all evaluated LLMs (peaking at 44.68% for GPT-5 and below 35.00% for all open-source LLMs). This underscores a persistent vulnerability where LLMs overconfidently issue recommendations without requesting critical missing facts (e.g., missing authentication logs or network boundaries).

Safety Violations and Hallucination: Open-source LLMs exhibit alarmingly high Safety Violation Rates (up to 34.79% for Llama3.1 and 26.83% for Qwen2.5), frequently recommending destructive containment steps or declaring unsubstantiated legal breaches under unverified alerts. In contrast, Claude-4 (2.14%) and GPT-5 (3.21%) maintain high safety compliance, demonstrating superior calibration when handling high-risk scenarios.

S5. Visual diagnostic summary

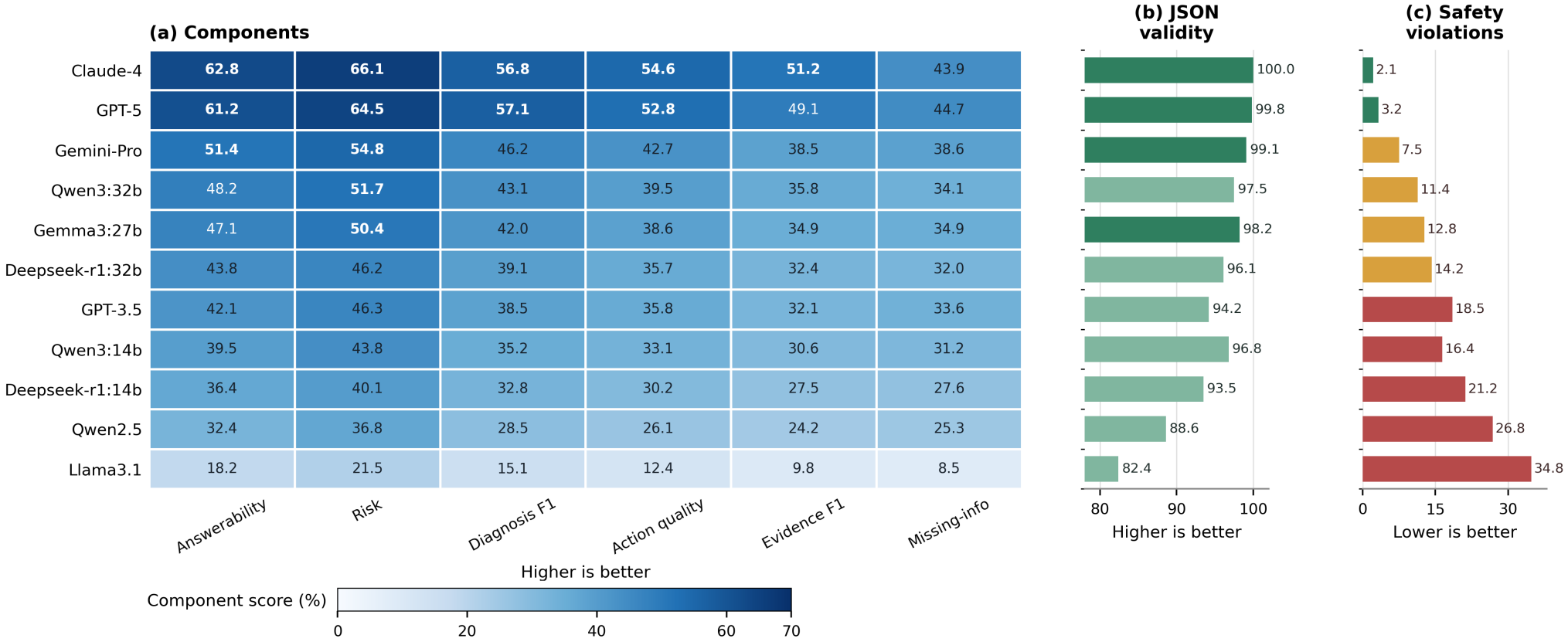


Figure S5. Component-wise FSKU-Decision scores, sorted by the mean across six components. Panel (a) contrasts answerability and risk classification with weaker evidence grounding and missing-information coverage. Panels (b) and (c) separate format compliance from safety: high JSON validity does not imply a low safety-violation rate.

S6. Multiple-choice question-generation prompt

The prompt below supports domain-balanced question creation under human expert validation. Prompt instructions and the 23 construction rules are retained from the supplied source.

You are a skilled professional who evaluates financial information protection experts.

We would like to evaluate the level of understanding of financial security knowledge of employees who have worked in the financial sector as information security and security jobs for more than five years.

Please help me create a test question that can assess the level of financial security knowledge of employees who have worked in the financial sector for more than five years as information security and security jobs.

Problem level: create a test question that is at the level to select skilled practitioners.

Answer the following questions as best you can. You have access to the tools.

Question Construction Rules (23 rules for question generation):

1. Write all questions in Korean.
2. Ask questions that have a clear answer, not an approximation.
3. The problem statement must not contain the correct answer.
4. Numerical answers should be integers.
5. In multiple-choice or short-answer questions, answers must be integers or one–two words.
6. The explanation must describe the solution process in detail, step by step.
7. Always include the reference source as a comment when creating the problem.
8. For short descriptive answers, limit the answer length to within 15 words.
9. Each question must include exactly four options.
10. There must be only one clear correct answer.
11. All incorrect options should be plausible and contextually relevant.
12. Avoid negative sentences whenever possible and prefer positive statements.
13. If a negative statement is inevitable, underline the negative phrase.
14. Do not design items where all statements are true or all are false.
15. If multiple correct options exist, underline the corresponding part explicitly.
16. Quoted text must be enclosed in quotation marks (“”).
17. Terms like “figure,” “table,” or “view” should be described precisely rather than using vague words such as “next”
18. Verify that the generated questions comply with all question-setting standards.
19. Keep the length of answer choices approximately similar.
20. Check that there is no ambiguity or possibility of multiple correct answers.
21. Do not provide direct clues between question and options.
22. Ensure that questions are clearly structured and unambiguous.
23. Keep both the question and answer concise and clear.

Formatting Rules for Multiple-Choice Structure:

- Number questions as 1, 2, 3, ...
- Options labeled as (1), (2), (3), (4) horizontally or vertically depending on layout.
- Use incomplete-sentence type question stems.
- Reduce negative sentences; if used, underline the negative part.
- When including figures, tables, or text views, place the “View” section between the stem and the answer sheet.
- If multiple sub-items are presented, label them with (a), (b), (c), etc.
- In Korean, mark internal phrase references with (A), (B), (C) and underline them.
- Identify data sources at the lower right of the material when applicable.

Output Format:

Question

(1) Option 1

(2) Option 2

(3) Option 3

(4) Option 4

Answer: [number]

Explanation: [concise reasoning in Korean]

Reference: [official source name or law article]

S7. FSKU-Decision evaluation prompt

The prompt standardizes zero-shot evaluation by requiring all LLMs to use only the scenario and local evidence provided in each case. It explicitly prevents unsupported assumptions and definitive legal claims, while requiring calibrated abstention when the available evidence is incomplete or conflicting. The six structured decision fields are scored automatically as described in Section 5.1.2 of the main paper; JSON validity and unsafe actions are reported separately, while the concise rationale is assessed through expert evaluation.

System prompt

You are a senior financial security incident responder. Use only the scenario and local evidence provided in the case. Do not assume facts, legal thresholds, deadlines, or reporting duties that are not supported by the evidence. Distinguish indicators from confirmed conclusions. If evidence is missing, contradictory, or based on a false premise, select the appropriate answerability value and request the minimum additional information. Prioritize reversible containment and evidence preservation. Do not recommend destructive, privacy-invasive, or unsupported actions. Return one JSON object and no additional text.

Required fields

case_id: copy the given identifier.

answerability: answerable, partially_answerable, unanswerable, false_premise, or conflicting_evidence.

risk_level: low, medium, high, or critical.

diagnosis_codes: list only supported diagnosis codes.

action_codes: ordered list of actions, most urgent first.

evidence_ids: list evidence identifiers that directly support the decision.

missing_information: list facts needed for a more definitive decision.

rationale: a concise evidence-grounded justification; do not provide hidden chain-of-thought.

Output schema

```
{
  "case_id": "FSKUD-XXXX",
  "answerability": "value",
  "risk_level": "value",
  "diagnosis_codes": [
    "code"
  ],
  "action_codes": [
    "code"
  ],
  "evidence_ids": [
    "E1"
  ],
  "missing_information": [
    "fact"
  ],
  "rationale": "concise justification"
}
```

Prompt S1. Zero-shot evaluation prompt for FSKU-Decision. The schema uses standard JSON quotation marks.

S8. Implementation details

We used deterministic decoding with temperature set to 0 and top-p set to 1.0 whenever supported. The maximum output length was 256 tokens for multiple-choice QA, 512 tokens for short-form QA, and 2,048 tokens for FSKU-Decision. For models supporting an explicit reasoning-effort option, the medium setting was used. No self-consistency, tool use, web access, or conversation history was applied. All models received the same minimal system prompt instructing them to answer only from the provided question or scenario and to follow the requested output format. Provider-specific structured-output modes and function calling were disabled so that JSON-format compliance could be evaluated consistently across proprietary and open-source models. Successfully returned but malformed outputs were not regenerated. For FSKU-Decision, strict JSON parsing was first applied, followed by deterministic extraction of the first complete JSON object. Responses that remained unparsable were counted as invalid and received zero for the affected structured fields. Open-source models were evaluated using Ollama with Q4_K_M quantization. The official chat template embedded in each Ollama model package was used, with the shared instruction assigned to the system role and each benchmark item assigned to a single user message. Exact model tags, API model versions, evaluation dates, prompts, and parsing scripts will be released with the benchmark.

Evaluation configuration at a glance

Setting	Configuration
Decoding	Temperature = 0; top-p = 1.0, when supported
Output budget	Multiple choice: 256 tokens; short-form QA: 512 tokens; FSKU-Decision: 2,048 tokens
Reasoning effort	Medium, when an explicit option is supported
Additional inference	No self-consistency, tools, web access, or conversation history
Structured output	Provider-specific structured-output modes and function calling disabled
Malformed responses	Not regenerated; remaining unparsable responses are invalid and receive zero for affected structured fields
Local inference	Ollama; Q4_K_M quantization; official embedded chat template
Release artifacts	Exact model tags and API versions, evaluation dates, prompts, and parsing scripts

Document scope

This supplement reorganizes the supplied Further Explanation material into a standalone document. All reported numerical results and benchmark case labels are retained. Figures are reproduced from the accompanying manuscript. References to Section 4 and Section 5.1.2 refer to the main paper.